

R0.73N | 固定成员有限应变稳定性与族转移障碍

R0.73N 对 R0.73M 的“变背景族能否转化为单个固定背景不稳定性”作出否定判定。对显式无外力双谐波背景族中的每个固定有限 Λ ，直接能量、交换子与延拓估计给出一个正的全三维同步 (H^3, H^3) 稳定管；R0.73M 所捕捉的是跨无界标记背景族的非一致指向放大，而不是任一固定成员处的 Lyapunov 不稳定性。

状态 · R0.73N 完成

Internal continuum theorem + claim-boundary audits

版本 v0.73N · 2026-08-31

full 3D synchronized (H_3, H_3) : INTERNAL THEOREM

planar synchronized (H_{3pl}, L_{2pl}) : INTERNAL THEOREM

full 3D FPS (H_3, L_2) : OPEN

FINITE / FIGURE: separate gates

NOT CLAY

00 / CANONICAL REPORT

Direct result

R0.73N closes a specific attempted inference from R0.73M. It does not turn the R0.73M family-level departure into instability of one background. It proves the opposite statement needed for that route: every fixed finite member of the explicit family has a positive full-three-dimensional H^3 stability tube.

On the normalized standard three-torus, with viscosity one and zero forcing, consider the mean-zero divergence-free trajectory

$$\bar{U}_\Lambda(t, y) = \left(0, 0, -\Lambda e^{-4t} \sin 2y + \frac{\Lambda}{2} e^{-16t} \sin 4y \right), \quad 0 < \Lambda < \infty. \quad (1.1)$$

For every fixed Λ , the internal continuum proof establishes forward synchronized (H^3, H^3) stability from the fixed initial time $t_0 = 0$. This is a fixed-trajectory extension of the FPS norm-quantifier pattern, not an FPS theorem for a nonstationary orbit, not orbital stability, and not a claim uniform over all starting times.

The decisive route correction is:

The amplification in R0.73M becomes unbounded only while the marked background changes with Λ . At each individual marked background, the local chordal moduli are finite and the full-three-dimensional H^3 synchronized stability radius is positive.

Four topology statements that must remain separate

Phase space and topology	Initial smallness	Observed distance	Global solution requirement	R0.73N state
full 3D synchronized (H^3, H^3)	H^3	H^3	proved inside the fixed-member tube	CLOSED
full 3D custom H^3 -in/ L^2 -out	H^3	L^2	inherited from the H^3 tube	CLOSED AS COROLLARY
planar synchronized ($H^3_{\text{pl}}, L^2_{\text{pl}}$)	genuinely L^2_{pl} only, with H^3_{pl} regularity	L^2_{pl}	global by two-dimensional regularity	CLOSED
full 3D FPS (H^3, L^2)	genuinely L^2 only, with arbitrary H^3 size	L^2	would require global H^3 continuation	OPEN

The second row must not be renamed as the fourth. In the FPS convention, (H^3, L^2) means that the datum is H^3 -regular but small only in L^2 ; Ro.73N does not prove that full-three-dimensional statement.

Exact relative-energy estimate

Let V be any strong solution of the same equation on $[0, T]$, and set $w = V - \bar{U}_\Lambda$. The perturbation equation gives the exact relative-energy identity

$$\frac{1}{2} \frac{d}{dt} \|w\|_2^2 + \|\nabla w\|_2^2 = - \int_{\mathbb{T}^3} (\partial_y F_\Lambda) w_2 w_3 \, dx, \tag{3.1}$$

where F_Λ is the third component of (1.1). Since

$$\frac{1}{2} \|\partial_y F_\Lambda(t)\|_\infty \leq \Lambda(e^{-4t} + e^{-16t}),$$

Gronwall yields

$$\|w(T)\|_2 \leq e^{\Lambda j(T)} \|w(0)\|_2,$$

$$j(T) = \frac{1 - e^{-4T}}{4} + \frac{1 - e^{-16T}}{16}.$$

$$\tag{3.2}$$

Because $j(T) \uparrow 5/16$,

$$\sup_{0 \leq t < T_{\max}} \|w(t)\|_2 \leq e^{5\Lambda/16} \|w(0)\|_2 \tag{3.3}$$

on every common strong lifespan. Equation (3.3) is all-time only when the comparison solution is known to be global: in the invariant planar subsystem or inside the full-three-dimensional tube proved below.

03 / CANONICAL REPORT

The genuine planar L^2 -small theorem

The subspace

$$\mathcal{S}_{2D} = \{(0, v_2(y, z), v_3(y, z)) : \partial_y v_2 + \partial_z v_3 = 0\} \quad (4.1)$$

is exactly invariant. Every H_{pl}^3 datum in it produces a global two-dimensional strong solution. Hence, for any $\epsilon > 0$,

$$\|w(0)\|_2 < \epsilon e^{-5\Lambda/16} \implies \sup_{t \geq 0} \|w(t)\|_2 < \epsilon. \quad (4.2)$$

The smallness assumption in (4.2) is genuinely L^2 , not H^3 . This is planar synchronized $(H_{\text{pl}}^3, L_{\text{pl}}^2)$ stability, with the regularity and distance norms playing the FPS roles but with a fixed nonstationary trajectory rather than an equilibrium.

04 / CANONICAL REPORT

Positive full-three-dimensional H^3 tube

Use the normalized Bessel-potential norm and set

$$X(t) = \|w(t)\|_{H^3}^2, \quad Y(t) = \|\nabla w(t)\|_{H^3}^2.$$

Mean zero gives $X \leq Y$. Periodic Kato--Ponce/Moser estimates and the top-order transport cancellations give a universal $C_3 \geq 1$ such that

$$\frac{1}{2}X'(t) + Y(t) \leq C_3 \|\bar{U}_\Lambda(t)\|_{H^4} X(t) + C_3 X(t)^{1/2} Y(t). \quad (5.1)$$

The two heat modes satisfy

$$A_{4,\Lambda} := \int_0^\infty \|\bar{U}_\Lambda(t)\|_{H^4} dt \leq \frac{489}{32\sqrt{2}} \Lambda. \quad (5.2)$$

Taking

$$r_3 = \frac{1}{4C_3}, \quad R_\Lambda = r_3 e^{-C_3 A_{4,\Lambda}}, \quad (5.3)$$

the bootstrap closes for every datum satisfying $\|w(0)\|_{H^3} < R_\Lambda$. Strong continuation then gives a unique global solution and

$$\sup_{t \geq 0} \|w(t)\|_{H^3} \leq e^{C_3 A_{4,\Lambda}} \|w(0)\|_{H^3}. \quad (5.4)$$

For every $\epsilon > 0$, one sufficient synchronized-stability radius is

$$\delta_\Lambda(\epsilon) = \min\{R_\Lambda, \epsilon e^{-C_3 A_{4,\Lambda}}\}. \quad (5.5)$$

The radius may become exponentially small as Λ grows, but it is strictly positive for every fixed finite member. It is a sufficient radius, not an optimal threshold.

Equation (5.4) also gives the custom full-three-dimensional H^3 -in/ L^2 -out corollary through $\|w\|_2 \leq \|w\|_{H^3}$. It does not close full-three-dimensional FPS (H^3, L^2) stability.

05 / CANONICAL REPORT

What Ro.73M actually implies for the flow map

For a fixed observation time T , let

$$\mathcal{D}_T = \{u_0 \in H_{\sigma,0}^3 : \text{the unique strong solution from } u_0 \text{ exists on } [0, T]\} \quad (6.1)$$

and let $S_T : \mathcal{D}_T \rightarrow H_{\sigma,0}^3$ be the time- T state map. The local chordal moduli below are localized by H^3 neighborhoods; the superscript $2 \rightarrow 2$ records only the norms in its quotient.

Precisely,

$$\mathfrak{L}_{\Lambda,\text{loc}}^{2 \rightarrow 2}(T) := \limsup_{r \downarrow 0} \sup_{\substack{h \in C_{\sigma,0}^\infty, 0 < \|h\|_{H^3} < r \\ \bar{U}_\Lambda(0) + h \in \mathcal{D}_T}} \frac{\|S_T(\bar{U}_\Lambda(0) + h) - S_T(\bar{U}_\Lambda(0))\|_2}{\|h\|_2}. \quad (6.2)$$

The modulus $\mathfrak{L}_{\Lambda,\text{loc}}^{3 \rightarrow 2}(T)$ uses the same formula with $\|h\|_{H^3}$ in the denominator.

At

$$D_* = \frac{1}{450}, \quad T_* = \frac{1}{1800}, \quad \mathcal{A}_* = \int_0^{D_*} \lambda_0(d) \, dd,$$

Ro.73M and the energy upper bound imply, for every sufficiently large Λ ,

$$c_* e^{\Lambda \mathcal{A}_*} \leq \mathfrak{L}_{\Lambda, \text{loc}}^{2 \rightarrow 2}(T_*) \leq e^{\Lambda j_*}. \quad (6.3)$$

The launch estimate $\|\phi_\Lambda\|_{H^3} \leq C_H \Lambda^2$ gives the mixed-topology bracket

$$\frac{c_*}{C_H} \Lambda^{-2} e^{\Lambda \mathcal{A}_*} \leq \mathfrak{L}_{\Lambda, \text{loc}}^{3 \rightarrow 2}(T_*) \leq e^{\Lambda j_*}. \quad (6.4)$$

The exponents are compatible because

$$j_* > \frac{359}{324000} > \frac{173}{450000} > \mathcal{A}_*. \quad (6.5)$$

The first strict inequality follows analytically from $1 - e^{-x} > x - x^2/2$; it is not inferred from decimal evaluation. Equations (6.3)–(6.4) do not identify a sharp exponent or a prefactor limit.

At any fixed $0 < \rho \leq \rho_0$, the Ro.73M marked pairs satisfy

$$\|h_{\Lambda, \rho}\|_{H^3} \leq C_H \rho \Lambda^2 e^{-\Lambda \mathcal{A}_*} \longrightarrow 0,$$

while their time- T_* output separation is at least $c_* \rho$. Define

$$\mathcal{E}_\rho = \{\bar{U}_\Lambda(0), \bar{U}_\Lambda(0) + h_{\Lambda, \rho} : \Lambda \geq \Lambda_0\} \subset \mathcal{D}_{T_*}. \quad (6.6)$$

The single Navier–Stokes flow map S_{T_*} fails to be uniformly continuous from H^3 to L^2 on this explicit unbounded planar set of marked pairs. Equivalently, the pointed maps fail to be equicontinuous across the family. Every individual marked basepoint still has a finite local modulus.

This is one flow map at different basepoints, not a different solution map for each Λ .

06 / CANONICAL REPORT

The quantifier obstruction

Ro.73M has the form

$$\exists \rho_0, c_*, \Lambda_0 \quad \forall \Lambda \geq \Lambda_0 \quad \forall \rho \in (0, \rho_0],$$
(7.1)

where the base is $\overline{U}_\Lambda$. With ρ fixed, $\Lambda \rightarrow \infty$ makes the selected input tend to zero and retains an order- ρ endpoint, but it also changes the background.

Fix instead one admissible $\Lambda^\sharp \geq \Lambda_0$. When $\rho \downarrow 0$, the input tends to zero and the relative upper bound forces the actual endpoint distance to zero. Sections 3--5 give a positive stability radius for every fixed $\Lambda > 0$, including members below the Ro.73M threshold. Hence the family-level quantifiers cannot be exchanged for arbitrarily small H^3 input and fixed L^2 escape at one member.

07 / CANONICAL REPORT

Why exact symmetry and compactness do not transfer the family

The registered candidates fail for explicit algebraic reasons.

Candidate	What it preserves or changes	Decision for this route
pure amplitude multiplication	preserves the shear only because its self-advection vanishes; changes perturbed nonlinear dynamics	not a neighborhood symmetry
time translation	must match heat rates 4 and 16 simultaneously	only the zero shift preserves both modes
spatial translation or Galilean change	changes phase or adds a zero mode	cannot change Λ
parabolic scaling	changes torus representation, Fourier rows, time, and H^3 size unless scale is one	no fixed-torus conjugacy
original-time compactness	$\ \overline{U}_\Lambda(0)\ _2^2 = 5\Lambda^2/8$	no bounded Sobolev subsequence
time-shift compactness	every bounded limit loses the second harmonic	loses the certified mechanism
smooth infinite-block embedding	fixed smooth heat shears retain finite integrated H^4 size	remains inside the finite-strain tube class

This audit closes only the registered transfer routes. A different forced, nondecaying, infinite-strain, or otherwise structurally different background would require a new problem and a new theorem.

08 / CANONICAL REPORT

Literature position

The bounded two-wave source audit fixes three boundaries.

1. Friedlander--Pavlović--Shvydkoy define the (X, Z) norm roles for a fixed equilibrium and prove an autonomous spectral-to-nonlinear theorem; they do not prove the present nonstationary synchronized theorem.
2. Grenier-type constructions, Couette threshold results, and classical non-normal transient growth show that parameter-dependent large gain is a meaningful phenomenon, but do not exchange a varying-background family for one fixed member.
3. Heat-evolving and periodic-shear stability or frozen-spectrum results are neighboring comparisons with different geometry, hypotheses, or conclusions. They are not provenance for the internal Ro.73N tube.

Primary links and exact support classes are recorded in `r073n_literature_audit.md` and `r073n_claim_source_ledger.md`. The strongest permitted absence statement is:

No single source in the recorded bounded search supplies the Ro.73N fixed-member theorem and family-transfer obstruction as a black box.

This is not an exhaustive classification and carries no novelty, first-result, or priority claim.

09 / CANONICAL REPORT

Evidence ledger

Evidence layer	State	Exact support
fixed-time relative L^2 bound	CLOSED	direct nonlinear integration by parts and Gronwall
finite all-time strain envelope	CLOSED	exact two-mode heat formula
planar synchronized $(H^3_{\text{pl}}, L^2_{\text{pl}})$ stability	CLOSED	planar invariance, global 2D regularity, and relative L^2 bound
full-3D synchronized (H^3, H^3) stability	CLOSED	periodic commutator estimate, finite H^4 integral, bootstrap, and continuation
custom full-3D H^3 -in/ L^2 -out	CLOSED AS COROLLARY	H^3 theorem plus $H^3 \hookrightarrow L^2$
full-3D FPS (H^3, L^2) stability	OPEN	no theorem for arbitrary H^3 -regular data small only in L^2
pointed family sensitivity and marked-set nonuniform continuity	CLOSED	sealed Ro.73M lower theorem plus Ro.73N upper estimate
registered symmetry and compactness transfer routes	CLOSED AS OBSTRUCTION	direct transformation and Sobolev/Fourier bookkeeping
independent analytic audit	PASS	complete rederivation of estimates, quantifiers, moduli, and transformations
adversarial audit	PASS	factor-of-two, lifespan, topology, bootstrap, modulus, and same-map attacks
bounded literature boundary	PASS	two-wave primary-source audit and internal claim-boundary reconciliation
finite diagnostic	CLOSED / PASS	19-file package; primary, independent, and 19-check final validation; sealed to the immutable source commit
formal figure	PASS	25-file package; PDF/SVG/600-dpi PNG, 605 source rows, 12 programmatic checks, and visual QA
public release content	READY	report, dictionary, figure, and claim-state vocabulary are frozen; deployment remains a separate transactional gate

10 / CANONICAL REPORT

Sealed finite diagnostic

The current Ro.73N diagnostic package has passed its primary and independent checks as a theorem-relevant error-detection layer. Its 9 source files are byte-bound to source commit 23526aa667aba1880a768ff4c1db35536edce737; the 19-file package has `sourceCommitAssigned=true`, `finalSeal=true`, and a passing 19-check final validator. Its checks cover:

- the exact normalized strain envelope $e^{-4t} + e^{-16t}$, including its equality witness;
- the closed form for $j(T)$ and $j(\infty) = 5/16$;
- the high-precision value $j_* = 0.0011080324480805907920035217032737848 \dots$;
- the exact strict rational chain in (6.5);

- illustrative exponent-factor curves across the marked background family.

The bound $\mathcal{A}_* < 173/450000$ is inherited from the sealed Ro.73M continuum theorem and is not recomputed by the finite package.

The sealed source-data file has exactly 605 rows: 241 strain samples, 243 cumulative samples, and 121 marked-basepoint samples. The primary validation and the independent Decimal/Fraction reconstruction both report `allChecksPass=true`; the latter evaluates six sentinel times without importing the primary producer.

These facts are finite checks, not proof of the continuum energy inequality, the H^3 stability tube, the inherited action interval, a sharp flow-map modulus, or any open statement.

11 / CANONICAL REPORT

Formal journal figure

Status: PASS

The frozen figure identifier is `fig-r073n-finite-strain-bracket`. Its 25-file archive contains 10 source files and 15 generated files, including a row-for-row binding to all 605 certificate rows, PDF/SVG/600-dpi PNG masters, code, environment records, monitoring logs, hashes, and a QA report. The figure sources and upstream certificate are sealed to the same immutable commit. Twelve programmatic checks pass; color, grayscale, final-size, and independently rendered PDF surfaces were visually inspected and found legible and unclipped. The figure evaluates exact formulas at different marked basepoints and is not a measured sharp modulus or a fixed-background instability certificate.

12 / CANONICAL REPORT

Public release copy (zh)

Title

Ro.73N | 固定成员有限应变稳定性与族转移障碍

Lead

Ro.73N 对 Ro.73M 的“变背景族能否转化为单个固定背景不稳定性”作出否定 判定。对显式无外力双谐波背景族中的每个固定有限 Λ ，直接能量、 交换子与延拓估计给出一个正的全三维同步 (H^3, H^3) 稳定管；Ro.73M 所捕捉的是跨无界标记背景族的非一致指向放大，而不是任一固定成员处的 Lyapunov 不稳定性。

Home

本节必须同时保留四条不同结论：全三维同步 (H^3, H^3) 稳定性已经 闭合；全三维 H^3 -输入/ L^2 -输出只是前者的自定义推论；平面不变子系统内的同步 $(H^3_{\text{pl}}, L^2_{\text{pl}})$ 稳定性具有真正的 L^2 初值小性；全三维 FPS (H^3, L^2) 稳定性仍为 OPEN。这里的同步比较 固定从 $t_0 = 0$ 出发，不允许轨道相移，也不声称对所有起始时刻一致。

Recap

Ro.73M 证明：当 $\Lambda \rightarrow \infty$ 且背景随之改变时，按完整作用量 缩小的平面扰动可以在固定时刻保持固定阶输出。Ro.73N 补上固定成员 审计：任意共存强解满足有限累积应变控制；平面扰动全局满足 L^2 稳定估计；足够小的全三维 H^3 扰动落在正的全局强解稳定管内。

因此，Ro.73M 中的量词顺序不能交换为一个固定背景上的任意小输入、固定距离逃逸。

Literature

有界两轮原始来源检索核对了 FPS 的范数量词、自治谱不稳定转移、Grenier 型参数族构造、Couette 阈值与瞬态增长、热演化剪切流和周期衰减流。检索 中没有一篇来源可以作为 Ro.73N 固定成员定理与族转移障碍的黑箱证明。这一表述只针对已记录的检索范围，不是穷尽性、首创性或优先权声明。

Next

Ro.73N 的证书、正式附图与发布内容已经完成封存；公开页面仍须经过事务式生成、翻译、同步 PDF、全站测试和线上字节一致性检查。数学上的下一发布门是 Ro.73O：为强迫、非衰减或无限累积应变等结构不同的固定背景重新冻结候选问题，先寻找 能够逃离本节有限总应变稳定管的必要结构，而不能把 Ro.73M 的现有背景族直接 重新标记。

13 / CANONICAL REPORT

Exact public boundary

The public note may state:

Every fixed member of the explicit unforced two-harmonic family has a positive full-three-dimensional forward synchronized (H^3, H^3) stability tube from $t_0 = 0$. Ro.73M instead detects nonuniform pointed amplification across an unbounded family of marked backgrounds. The varying-background theorem therefore cannot be converted into instability of one member by the audited quantifier, symmetry, or compactness routes.

It must also state all of the following:

- the full-3D H^3 -in/ L^2 -out statement is a custom corollary;
- the planar $(H^3_{\text{pl}}, L^2_{\text{pl}})$ theorem has genuine L^2 initial smallness but applies only in the invariant planar subsystem;
- full-three-dimensional FPS (H^3, L^2) stability remains OPEN;
- the stability radius and both local-modulus exponent bounds are not claimed sharp;
- the same autonomous Navier--Stokes flow map is evaluated at different marked basepoints;
- the obstruction is specific to the registered Ro.73M transfer routes;
- finite diagnostics do not certify continuum claims;
- arbitrary-background behavior, transverse critical-norm growth, finite-time singularity, and the Clay problem remain OPEN.

The exact public boundary label is NOT CLAY.

The public note must not say that Ro.73M proves fixed-member Lyapunov instability, that large finite amplification is instability, that the full-three-dimensional FPS (H^3, L^2) problem is closed, or that the bounded source audit establishes novelty or priority.

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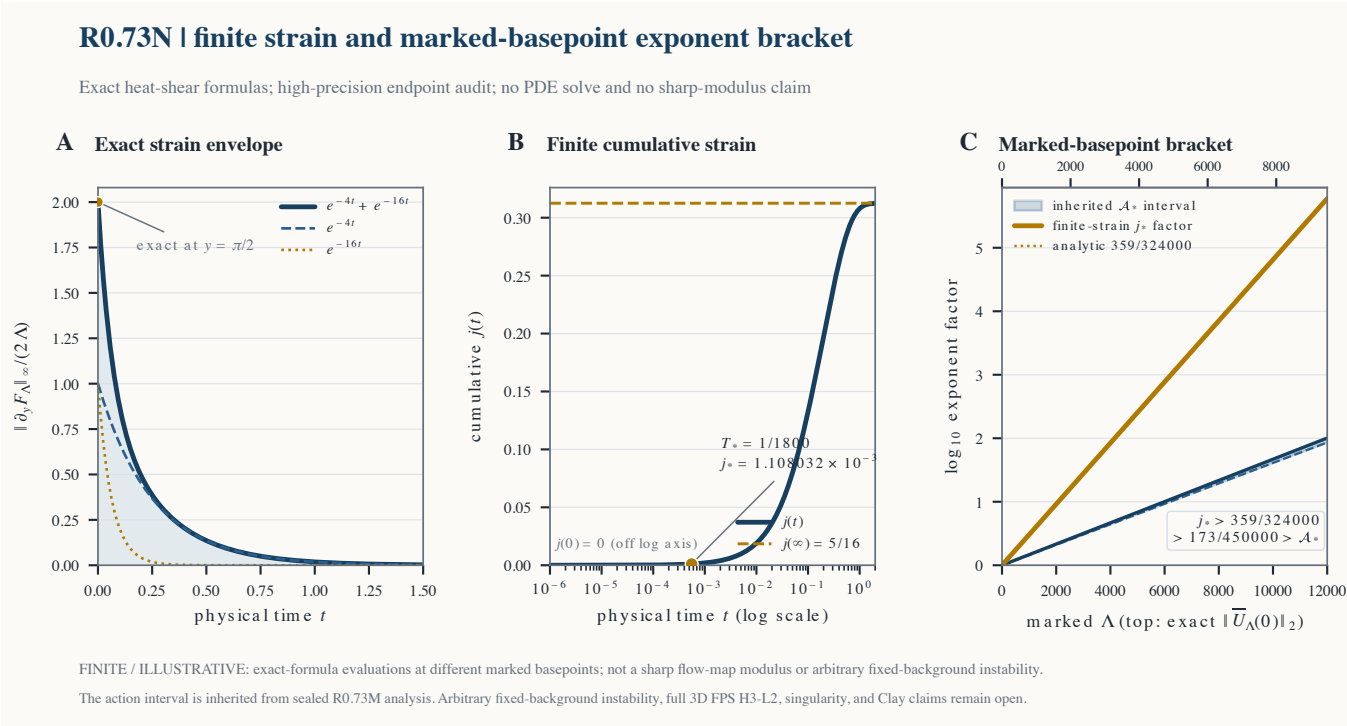
Canonical machine-readable boundary

fixedTimeRelativeL2LipschitzBound=CLOSED
finiteAllTimeStrainEnvelope=CLOSED
fixedMemberPlanarL2SynchronizedStability=CLOSED
fixedMemberThreeDimensionalH3SynchronizedStability=CLOSED
fullThreeDimensionalH3InputL2Output=CLOSED_AS_COROLLARY

familyFlowMapNonuniformMarkedBasepointSensitivity=CLOSED
familyDepartureImpliesFixedMemberInstability=FALSE_AS_INFERENCE
singleR073mMemberH3SmallL2FixedDistanceEscape=FALSE
amplitudeOnlyIdentificationIsNSSymmetry=FALSE
timeTranslationIdentifiesLambdaFamily=FALSE
parabolicScalingIdentifiesLambdaFamilyOnFixedTorus=FALSE
fullThreeDimensionalFPSH3L2Stability=OPEN
optimalFixedMemberStabilityRadius=OPEN
sharpFamilyLipschitzExponent=OPEN
arbitraryFixedBackgroundInstability=OPEN
transverseCriticalNormGrowth=OPEN
finiteTimeSingularity=OPEN
Clay=OPEN
finiteDiagnosticValidation=PASS
finiteDiagnosticPackage=CLOSED
sourceCommitAssigned=TRUE
finalSeal=TRUE
formalFigurePackage=PASS
publicReleaseContent=READY

F / JOURNAL FIGURE

有限应变包络、累积系数与标记基点指数括号



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B / EXACT RELEASE BOUNDARY

固定成员、平面子系统与开放拓扑严格分列

fixedTimeRelativeL2LipschitzBound=CLOSED; finiteAllTimeStrainEnvelope=CLOSED;
fixedMemberPlanarL2SynchronizedStability=CLOSED; fixedMemberThreeDimensionalH3SynchronizedStability=CLOSED;

fullThreeDimensionalH3InputL2Output=CLOSED_AS_COROLLARY;
familyFlowMapNonuniformMarkedBasepointSensitivity=CLOSED; finiteDiagnosticPackage=CLOSED

finiteAllTimeStrainEnvelope=CLOSED; finiteTimeSingularity=OPEN; finiteDiagnosticValidation=PASS;
finiteDiagnosticPackage=CLOSED; sourceCommitAssigned=TRUE; finalSeal=TRUE; formalFigurePackage=PASS;
publicReleaseContent=READY

familyDepartureImpliesFixedMemberInstability=FALSE_AS_INFERENCE;
singleRo73mMemberH3SmallL2FixedDistanceEscape=FALSE; fullThreeDimensionalFPSH3L2Stability=OPEN;
amplitudeOnlyIdentificationIsNSSymmetry=FALSE; timeTranslationIdentifiesLambdaFamily=FALSE;
parabolicScalingIdentifiesLambdaFamilyOnFixedTorus=FALSE; optimalFixedMemberStabilityRadius=OPEN;
sharpFamilyLipschitzExponent=OPEN; arbitraryFixedBackgroundInstability=OPEN; transverseCriticalNormGrowth=OPEN;
finiteTimeSingularity=OPEN; Clay=OPEN; originalTimeCompactness=FALSE;
boundedTimeShiftRetainsTwoHarmonics=FALSE; infiniteSmoothHeatShearEvadesFiniteStrainTube=FALSE;
differentForcedOrNondecayingBackgroundRoute=OPEN

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R / REPRODUCTION

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